

Bayesian Methods

Synopsis

Following a review of the classical least squares approach to solving inverse problems, we introduce the Bayesian approach, which treats the model as a random variable with a probability distribution that we seek to estimate. A prior distribution for the model parameters is combined with the data to produce a posterior distribution for the model parameters. In special cases, the Bayesian approach produces solutions that are equivalent to the least squares, maximum likelihood, and Tikhonov regularization solutions. Several examples of the Bayesian approach are presented. Markov Chain Monte Carlo methods for sampling from the posterior distribution are presented and demonstrated.

11.1. REVIEW OF THE CLASSICAL APPROACH

In the classical approach to parameter estimation and inverse problems with discrete data and models, we begin with a mathematical model of the form $\mathbf{G}\mathbf{m} = \mathbf{d}$ in the linear case or $G(\mathbf{m}) = \mathbf{d}$ in the nonlinear case. We assume that there exists a true model, $\mathbf{m}_{\text{true}}$, and a true data set, $\mathbf{d}_{\text{true}}$, such that $\mathbf{G}\mathbf{m}_{\text{true}} = \mathbf{d}_{\text{true}}$. We acquire an actual data set, $\mathbf{d}$, which is generally the sum of $\mathbf{d}_{\text{true}}$ and measurement noise. Our goal is to recover $\mathbf{m}_{\text{true}}$ from the noisy data.

For well-conditioned linear problems, under the assumption of independent and normally distributed data errors, the theory is well developed. In Chapter 2 it was shown that the maximum likelihood principle leads to the least squares solution, which is found by minimizing the 2-norm of the residual, $\|\mathbf{G}\mathbf{m} - \mathbf{d}\|_2$. Since there is noise in the data, we should expect some misfit between the data predictions of the forward model and the data, so that observed values of the square of the 2-norm of the inverse standard deviation-weighted residual, χ_{obs}^2 , will not typically be zero. We saw that the χ^2 distribution can be used to test the goodness-of-fit of a least squares solution. We showed that the least squares solution, $\mathbf{m}_{L_2}$, is an unbiased estimate of $\mathbf{m}_{\text{true}}$. We were also able to compute a covariance matrix for the estimated parameters

$$\text{Cov}(\mathbf{m}_{L_2}) = (\mathbf{G}^T \mathbf{G})^{-1} \mathbf{G}^T \text{Cov}(\mathbf{d}) \mathbf{G} (\mathbf{G}^T \mathbf{G})^{-1} \quad (11.1)$$

to compute confidence intervals for and correlations between the estimated parameters.

This approach works very well for linear regression problems where the least squares problem is well-conditioned. We found, however, that in many cases the least squares problem is not well-conditioned. In such situations, the set of solutions that adequately fits the data is large and diverse, and commonly contains physically unreasonable models.

In Chapters 4 through 8, we discussed a number of approaches to regularizing the least squares problem. These approaches pick a single “best” solution out of those that adequately fit the data, based on a preference for what sort of model features constitute a good solution. Zeroth-order Tikhonov regularization selects the model that minimizes the model 2-norm $\|\mathbf{m}\|_2$ subject to the residual norm constraint, $\|\mathbf{G}\mathbf{m} - \mathbf{d}\|_2 < \delta$, while higher-order Tikhonov regularization selects the model that minimizes a model seminorm $\|\mathbf{L}\mathbf{m}\|_2$ subject to $\|\mathbf{G}\mathbf{m} - \mathbf{d}\|_2 < \delta$. We introduced L_1 regularization and the related technique of total variation regularization which are now widely used in compressive sensing.

For relatively small linear problems, straightforward, insightful, and robust computation of regularized solutions can be performed with the help of singular value decomposition (SVD). For large sparse linear problems, iterative methods such as conjugate gradient least squares (CGLS) or LSQR are widely used.

For nonlinear problems, as discussed in Chapters 9 and 10, the Gauss-Newton, Levenberg-Marquardt, or Occam’s inversion methods can be used to find local minima of the nonlinear least squares problem. We showed how approximate confidence intervals for the fitted parameters can be obtained by linearizing the nonlinear model around the best fit parameters. As in linear inverse problems, the nonlinear least squares problem can be badly conditioned in which case regularization may be needed to obtain a stable solution.

Unfortunately nonlinear problems may have a large number of local minimum solutions, and finding the global minimum can be difficult. Furthermore, if there are several local minimum solutions with acceptable data fits, then it may be difficult to select a single “best” solution.

How can we more generally justify selecting one solution from the set of models that adequately fit the data? One justification is Occam’s razor, which is the philosophy that when we have several different hypotheses to consider, we should select the simplest. Solutions selected by regularization are in some sense the simplest models that fit the data. However, this approach is not by itself entirely satisfactory because different choices of the regularization term used in obtaining regularized solutions can result in very different models, and the specific choice of regularization may be subjective.

Recall from Chapter 4 (e.g., Example 4.3) that once we have regularized a least squares problem, we lose the ability to obtain statistically useful confidence intervals for the parameters because regularization introduces bias. In particular, the expected value of the regularized solution is not generally equal to the true model. In practice

this regularization bias is often much more significant than the effect of noise in the data. Bounds on the error in Tikhonov regularized solution error were discussed in Section 4.8, but these estimates require knowledge of the true model that is typically not available in practice.

11.2. THE BAYESIAN APPROACH

The Bayesian approach is named after Thomas Bayes, an 18th-century pioneer in probability theory. The methodology is based on philosophically different ideas than we have considered so far. However, as we will see, it frequently results in similar solutions.

The most fundamental difference between the classical and Bayesian approaches is in the conceptualization of the solution. In the classical approach, there is a specific but unknown model $\mathbf{m}_{\text{true}}$ that we would like to uncover. In the Bayesian approach the model is not deterministic, but is rather a random variable, and the solution takes the form of a probability distribution for the model parameters called the **posterior distribution**. Once we have this probability distribution, we can use it to answer probabilistic questions about the model, such as “What is the probability that m_5 is less than 1?” In the classical approach such questions do not make sense, since the true model that we seek is not a random variable.

A second very important difference between the classical and Bayesian approaches is that the Bayesian approach naturally incorporates prior information about the solution, ranging from hard additional constraints to experience-based intuition. This information is expressed mathematically as a **prior distribution** for the model. Once data have been collected, they are combined with the prior distribution using Bayes’ theorem (B.54) to produce the desired posterior distribution for the model parameters.

If no other information is available, then under the **principle of indifference**, we may alternatively pick a prior distribution where all model parameter values have equal likelihood. Such a prior distribution is said to be **uninformative**.

It should be pointed out that, in the common case where the model parameters are contained in the range $(-\infty, \infty)$, the uninformative prior is not a proper probability distribution. This is because a probability density function $f(x)$ does not exist which satisfies (B.4) so that

$$\int_{-\infty}^{\infty} f(x) \, dx = 1 \quad (11.2)$$

and $f(x)$ is constant. In practice, the use of this improper prior distribution in Bayesian methods can nevertheless be justified because the resulting posterior distribution for the model is a proper distribution.

One of the main objections to the Bayesian approach is that the method is “unscientific” because it allows the analyst to incorporate subjective judgments into the model that are not solely based on the data. Proponents of the approach rejoin that there are also subjective aspects to the classical approach embodied in the choice of regularization biases, and furthermore, that one is free to choose an uninformative prior distribution.

Here, we denote the prior distribution by $p(\mathbf{m})$, and assume that we can compute the conditional probability distribution, $f(\mathbf{d}|\mathbf{m})$, that, given a particular model, corresponding data, $\mathbf{d}$, will be observed. Given a prior distribution, we then seek the conditional (posterior) distribution of the model parameter(s) given the data. We will denote this posterior probability distribution for the model parameters by $q(\mathbf{m}|\mathbf{d})$. Bayes’ theorem relates the prior and posterior distributions in a way that makes the computation of $q(\mathbf{m}|\mathbf{d})$ possible, and can be stated as follows.

Theorem 11.1

$$q(\mathbf{m}|\mathbf{d}) = \frac{f(\mathbf{d}|\mathbf{m})p(\mathbf{m})}{c} \quad (11.3)$$

where

$$c = \int_{\text{all models}} f(\mathbf{d}|\mathbf{m})p(\mathbf{m})d\mathbf{m}. \quad (11.4)$$

Note that the constant c in (11.3) simply normalizes the conditional distribution $q(\mathbf{m}|\mathbf{d})$ so that its integral in model space is one.

For some purposes, knowing the normalization constant, c , is not necessary. For example, we can compare two models $\hat{\mathbf{m}}$ and $\bar{\mathbf{m}}$ by computing the likelihood ratio:

$$\frac{q(\hat{\mathbf{m}}|\mathbf{d})}{q(\bar{\mathbf{m}}|\mathbf{d})} = \frac{f(\mathbf{d}|\hat{\mathbf{m}})p(\hat{\mathbf{m}})}{f(\mathbf{d}|\bar{\mathbf{m}})p(\bar{\mathbf{m}})}. \quad (11.5)$$

A very small likelihood ratio would indicate that the model $\bar{\mathbf{m}}$ is far more likely than the model $\hat{\mathbf{m}}$. Because c is not always needed, (11.3) is often written as a statement of proportionality,

$$q(\mathbf{m}|\mathbf{d}) \propto f(\mathbf{d}|\mathbf{m})p(\mathbf{m}). \quad (11.6)$$

However, there are many other situations in which knowing c in (11.3) is required. In particular, c is required to compute any posterior probabilities and to compute the expected value and variance of the posterior distribution.

It is important to re-emphasize that the probability distribution $q(\mathbf{m}|\mathbf{d})$ does not provide a single model that we can consider to be the “answer.” However, in cases where we want to single out a representative model, it may be appropriate to identify

the one corresponding to the largest value of $q(\mathbf{m}|\mathbf{d})$. This model is referred to as the **maximum a posteriori (MAP)** model. Another possibility is to select the mean of the posterior distribution. In situations where the posterior distribution is normal, the MAP and posterior mean models will be identical.

In general, the computation of a posterior distribution using (11.3) can be difficult. The chief difficulty lies in evaluating the integral in (11.4). This integral often has very high dimensionality, and numerical integration techniques may thus be computationally daunting.

Fortunately, there are a number of useful special cases in which the computation of the posterior distribution is greatly simplified. One simplification occurs when the prior distribution $p(\mathbf{m})$ is uninformative, in which case (11.6) simplifies to

$$q(\mathbf{m}|\mathbf{d}) \propto f(\mathbf{d}|\mathbf{m}), \quad (11.7)$$

and the posterior distribution is precisely the likelihood function, $L(\mathbf{m}|\mathbf{d})$. Under the maximum likelihood principle, we would select the model $\mathbf{m}_{\text{ML}}$ that maximizes $L(\mathbf{m}|\mathbf{d})$, which is the MAP model.

A further simplification occurs when data noise elements are independent and normally distributed with standard deviation σ . Because the data errors are independent, we can write the likelihood function as

$$L(\mathbf{m}|\mathbf{d}) = f(\mathbf{d}|\mathbf{m}) = f(d_1|\mathbf{m}) \cdot f(d_2|\mathbf{m}) \cdots f(d_n|\mathbf{m}). \quad (11.8)$$

If the data points d_i are normally distributed with expected values given by the (linear or nonlinear) operation $(G(\mathbf{m}))_i$, and each has standard deviation σ , we can write

$$f(d_i|\mathbf{m}) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{((G(\mathbf{m}))_i - d_i)^2}{2\sigma^2}} \quad (11.9)$$

and

$$L(\mathbf{m}|\mathbf{d}) = \left(\frac{1}{\sigma\sqrt{2\pi}} \right)^m e^{-\sum_{i=1}^m \frac{((G(\mathbf{m}))_i - d_i)^2}{2\sigma^2}}. \quad (11.10)$$

We can maximize (11.10) by maximizing the exponent or equivalently minimizing the negative of the exponent.

$$\min \sum_{i=1}^m \frac{((G(\mathbf{m}))_i - d_i)^2}{2\sigma^2}. \quad (11.11)$$

This is a weighted least squares problem. Thus we have shown that when we have independent and normally distributed measurement errors and we use an uninformative prior, the MAP solution is the least squares solution.



Example 11.1

Consider a very simple parameter estimation problem where we perform repeated weighings of a microscopic object to determine its mass in micrograms. The measurement errors are normally distributed with zero mean and standard deviation $\sigma = 1 \mu\text{g}$. Our goal is to estimate the mass of the object.

For the specified normally distributed and zero mean measurement error, the probability density function for a measurement d given m is

$$f(d|m) = \frac{1}{\sqrt{2\pi}} e^{-(m-d)^2/2}. \quad (11.12)$$

Suppose that we weigh the mass once and obtain a measurement of $d_1 = 10.3 \mu\text{g}$. What do we now know about m ? For an uninformative prior, (11.7) gives

$$q(m|d_1 = 10.3) \propto f(10.3|m) = \frac{1}{\sqrt{2\pi}} e^{-(m-10.3)^2/2}. \quad (11.13)$$

Because (11.13) is itself a normal probability distribution, the constant of proportionality in (11.3) is 1, and the posterior distribution for the mass in micrograms (Figure 11.1) is therefore

$$q(m|d_1 = 10.3) = \frac{1}{\sqrt{2\pi}} e^{-(m-10.3)^2/2}. \quad (11.14)$$

Next, suppose that we obtain a second statistically independent measurement of $d_2 = 10.1 \mu\text{g}$. We can then use the distribution (11.14) estimated from the first measurement

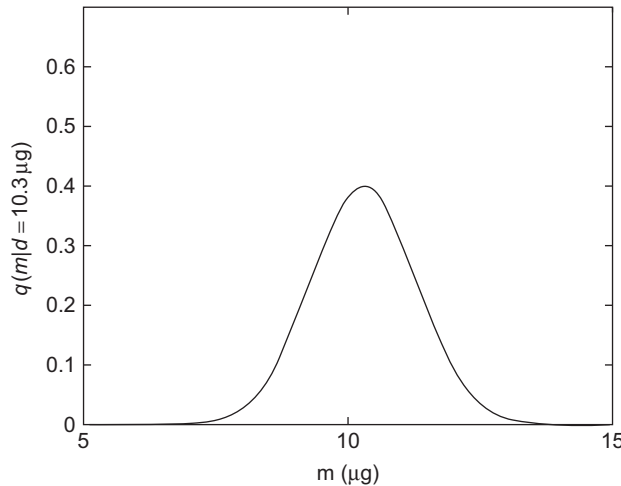


Figure 11.1 Posterior distribution $q(m|d_1 = 10.3 \mu\text{g})$, uninformative prior.

as an informative prior distribution to compute a revised posterior distribution:

$$\begin{aligned} q(m|d_1 = 10.3, d_2 = 10.1) &\propto f(d_2 = 10.1|m)q(m|d_1 = 10.3) \\ &= \frac{1}{\sqrt{2\pi}} e^{-(m-10.1)^2/2} \frac{1}{\sqrt{2\pi}} e^{-(m-10.3)^2/2}. \end{aligned} \quad (11.15)$$

Combining the exponents and absorbing the $1/\sqrt{2\pi}$ factors into the constant of proportionality gives

$$q(m|d_1 = 10.3, d_2 = 10.1) \propto e^{-((m-10.3)^2 + (m-10.1)^2)/2}. \quad (11.16)$$

Finally, we can simplify the exponent by combining terms and completing the square to obtain

$$(m - 10.3)^2 + (m - 10.1)^2 = 2(m - 10.2)^2 + 0.02. \quad (11.17)$$

Thus,

$$q(m|d_1 = 10.3 \text{ } \mu\text{g}, d_2 = 10.1) \propto e^{-(2(m-10.2)^2 + 0.02)/2}. \quad (11.18)$$

The constant $e^{-0.02/2}$ can be absorbed into the constant of proportionality, giving

$$q(m|d_1 = 10.3, d_2 = 10.1) \propto e^{-(10.2-m)^2}. \quad (11.19)$$

Normalizing (11.19) gives a normal posterior distribution

$$q(m|d_1 = 10.3, d_2 = 10.1) = \frac{1}{(1/\sqrt{2})\sqrt{2\pi}} e^{-\frac{(10.2-m)^2}{2(1/\sqrt{2})^2}}, \quad (11.20)$$

with mean 10.2 μg and $\sigma = 1/\sqrt{2}$ μg (Figure 11.2). Since we used an uninformative prior and the measurement errors were independent and normally distributed, the MAP solution is precisely the least squares solution for this problem.

It is notable in the second part of this example that we started with a normal prior distribution, accommodated normally distributed data, and obtained a normal posterior distribution (see (11.20)). In general we should not expect that the prior and posterior distributions will both be well-known distributions whose properties are well known. A prior distribution associated with a simple posterior distribution in this way is called a **conjugate prior**. There are other families of conjugate distributions for various parameter estimation problems, but in general this is unusual [51].

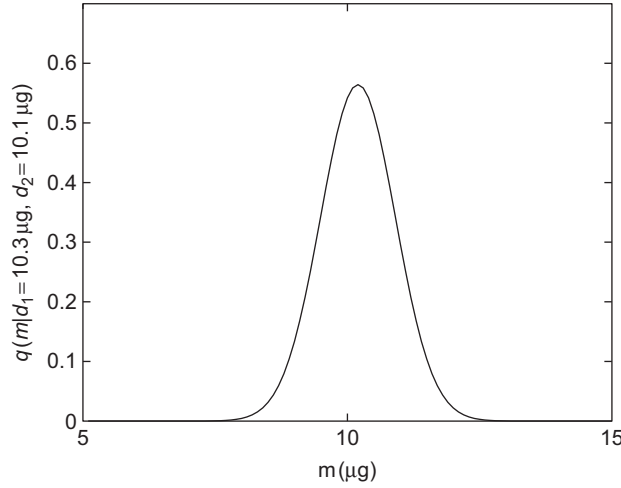


Figure 11.2 Posterior distribution $q(m|d_1 = 10.3 \mu\text{g}, d_2 = 10.1 \mu\text{g})$.

11.3. THE MULTIVARIATE NORMAL CASE

The result shown in [Example 11.1](#) in which a normal prior distribution and normally distributed data lead to a normal posterior distribution can be readily extended to problems with many model parameters. We next examine the problem of determining the posterior distribution for a linear model, multivariate normal (MVN) distributed data errors, and an MVN prior distribution.

Let $\mathbf{d}_{\text{obs}}$ be the observed data, $\mathbf{C}_D$ be the corresponding data covariance matrix, $\mathbf{m}_{\text{prior}}$ be the mean of the prior distribution, and $\mathbf{C}_M$ be the covariance matrix for the prior distribution. The prior distribution is thus, by (B.62),

$$p(\mathbf{m}) \propto e^{-\frac{1}{2}(\mathbf{m}-\mathbf{m}_{\text{prior}})^T \mathbf{C}_M^{-1}(\mathbf{m}-\mathbf{m}_{\text{prior}})}, \quad (11.21)$$

and the conditional distribution of the data, given $\mathbf{m}$, is

$$f(\mathbf{d}|\mathbf{m}) \propto e^{-\frac{1}{2}(\mathbf{Gm}-\mathbf{d})^T \mathbf{C}_D^{-1}(\mathbf{Gm}-\mathbf{d})}. \quad (11.22)$$

Thus, (11.6) gives

$$q(\mathbf{m}|\mathbf{d}) \propto e^{-\frac{1}{2}((\mathbf{Gm}-\mathbf{d})^T \mathbf{C}_D^{-1}(\mathbf{Gm}-\mathbf{d}) + (\mathbf{m}-\mathbf{m}_{\text{prior}})^T \mathbf{C}_M^{-1}(\mathbf{m}-\mathbf{m}_{\text{prior}}))}. \quad (11.23)$$

Tarantola [155] showed that this can be simplified to

$$q(\mathbf{m}|\mathbf{d}) \propto e^{-\frac{1}{2}(\mathbf{m}-\mathbf{m}_{\text{MAP}})^T \mathbf{C}_{M'}^{-1}(\mathbf{m}-\mathbf{m}_{\text{MAP}})}, \quad (11.24)$$

where $\mathbf{m}_{\text{MAP}}$ is the MAP solution, and

$$\mathbf{C}_{M'} = (\mathbf{G}^T \mathbf{C}_D^{-1} \mathbf{G} + \mathbf{C}_M^{-1})^{-1}. \quad (11.25)$$

The MAP solution can be found by maximizing the exponent in (11.23), or equivalently by minimizing its negative:

$$\min (\mathbf{G}\mathbf{m} - \mathbf{d})^T \mathbf{C}_D^{-1} (\mathbf{G}\mathbf{m} - \mathbf{d}) + (\mathbf{m} - \mathbf{m}_{\text{prior}})^T \mathbf{C}_M^{-1} (\mathbf{m} - \mathbf{m}_{\text{prior}}). \quad (11.26)$$

The key to minimizing (11.26) is to rewrite it in terms of the matrix square roots of $\mathbf{C}_M^{-1}$ and $\mathbf{C}_D^{-1}$. Note that every covariance matrix is positive definite and has a unique positive definite matrix square root, which may be calculated in MATLAB using the `sqrtm` routine. This minimization problem can then be reformulated as

$$\min (\mathbf{C}_D^{-1/2} (\mathbf{G}\mathbf{m} - \mathbf{d}))^T (\mathbf{C}_D^{-1/2} (\mathbf{G}\mathbf{m} - \mathbf{d})) + (\mathbf{C}_M^{-1/2} (\mathbf{m} - \mathbf{m}_{\text{prior}}))^T (\mathbf{C}_M^{-1/2} (\mathbf{m} - \mathbf{m}_{\text{prior}})), \quad (11.27)$$

or as the standard least squares problem,

$$\min \left\| \begin{bmatrix} \mathbf{C}_D^{-1/2} \mathbf{G} \\ \mathbf{C}_M^{-1/2} \end{bmatrix} \mathbf{m} - \begin{bmatrix} \mathbf{C}_D^{-1/2} \mathbf{d} \\ \mathbf{C}_M^{-1/2} \mathbf{m}_{\text{prior}} \end{bmatrix} \right\|_2^2. \quad (11.28)$$

Examining the right-hand terms in (11.28), note that

$$\text{Cov}(\mathbf{C}_D^{-1/2} \mathbf{d}) = \mathbf{C}_D^{-1/2} \mathbf{C}_D (\mathbf{C}_D^{-1/2})^T = \mathbf{I}. \quad (11.29)$$

The multiplication of $\mathbf{C}_D^{-1/2}$ times $\mathbf{d}$ in (11.28) can thus be conceptualized as a data transformation that both makes the resulting elements independent and normalizes the standard deviations. In the model space, multiplication by $\mathbf{C}_M^{-1/2}$ has the same effect.

An interesting limiting case is where the prior distribution provides essentially no information. Consider an MVN prior distribution with a covariance matrix $\mathbf{C}_M = \alpha^2 \mathbf{I}$, in the limit where α is extremely large. In this case, the diagonal elements of $\mathbf{C}_M^{-1}$ will be extremely small, and the posterior covariance matrix (11.25) will be well-approximated by

$$\mathbf{C}_{M'} \approx (\mathbf{G}^T \text{Cov}(\mathbf{d})^{-1} \mathbf{G})^{-1}. \quad (11.30)$$

If the data covariance matrix is $\mathbf{C}_D = \sigma^2 \mathbf{I}$, then

$$\mathbf{C}_{M'} \approx \sigma^2 (\mathbf{G}^T \mathbf{G})^{-1}, \quad (11.31)$$

which is precisely the covariance matrix for the least squares model parameters in (11.1). Furthermore, when we solve (11.28) to obtain the MAP solution, we find that it simplifies to the least squares problem of minimizing $\|\mathbf{G}\mathbf{m} - \mathbf{d}\|_2^2$. Thus, under the common assumption of normally distributed and independent data errors with constant variance,

a very broad prior distribution leads to a MAP solution that is the unregularized least squares solution.

It is also worthwhile to consider what happens in the special case where $\mathbf{C}_D = \sigma^2 \mathbf{I}$ and $\mathbf{C}_M = \alpha^2 \mathbf{I}$. In this case the corresponding matrix square roots are also proportional to identity matrices, and (11.28) simplifies to

$$\min (1/\sigma)^2 \|(\mathbf{G}\mathbf{m} - \mathbf{d})\|_2^2 + (1/\alpha)^2 \|\mathbf{m} - \mathbf{m}_{\text{prior}}\|_2^2, \quad (11.32)$$

which is a modified optimization problem for zeroth-order Tikhonov regularization (4.4), where the 2-norm regularization term is evaluated relative to $\mathbf{m}_{\text{prior}}$, and the equivalent Tikhonov regularization parameter is σ/α . Thus, the MAP solution obtained by using a prior with independent and normally distributed model parameters is precisely the zeroth-order Tikhonov regularized solution obtained by solving (11.32). However, this does not mean that the Bayesian approach is entirely equivalent to Tikhonov regularization, because the Bayesian solution is a probability distribution, whereas the Tikhonov solution is a single model from that distribution.

Once we have obtained the posterior distribution, it is straightforward to generate corresponding model realizations. We may wish to do this to assess likely or unlikely model features. Following the procedure outlined in Example B.10, we compute the Cholesky factorization of the posterior distribution covariance matrix,

$$\mathbf{C}_{M'} = \mathbf{R}^T \mathbf{R}, \quad (11.33)$$

and generate a random solution

$$\mathbf{m} = \mathbf{R}^T \mathbf{s} + \mathbf{m}_{\text{MAP}}, \quad (11.34)$$

where the vector $\mathbf{s}$ consists of independent and normally distributed random numbers with a zero mean and a unit standard deviation.

Example 11.2

We consider Bayesian solutions to the Shaw problem that was previously considered in Examples 3.3, 4.1, 4.2, 4.3, and 4.8.

We first use a relatively uninformative MVN prior distribution with mean 0.5, standard deviation 0.5, and zero covariances, so that $\mathbf{C}_M = 0.25 \mathbf{I}$. As in the previous examples, the measurement noise has standard deviation 1.0×10^{-6} , so that $\mathbf{C}_D = 1.0 \times 10^{-12} \mathbf{I}$. Solving (11.28) produces the $\mathbf{m}_{\text{MAP}}$ solution shown in Figure 11.3. Figure 11.4 shows this same solution with error bars. These error bars are not classical 95% confidence intervals (e.g., Figure 4.9). Rather, they are 95% probability intervals calculated from the MVN posterior distribution, so that there is 95% probability that each model parameter lies within the corresponding symmetric interval around $\mathbf{m}_{\text{MAP}}$.



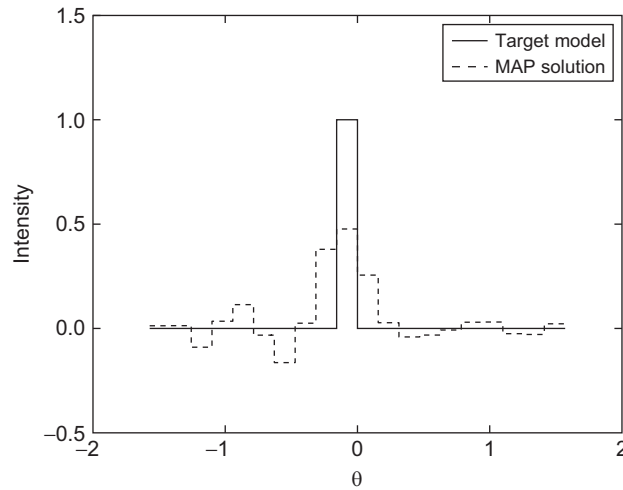


Figure 11.3 The MAP solution and the true model for the Shaw example using an MVN prior distribution with mean 0.5, standard deviation 0.5, and zero covariance.

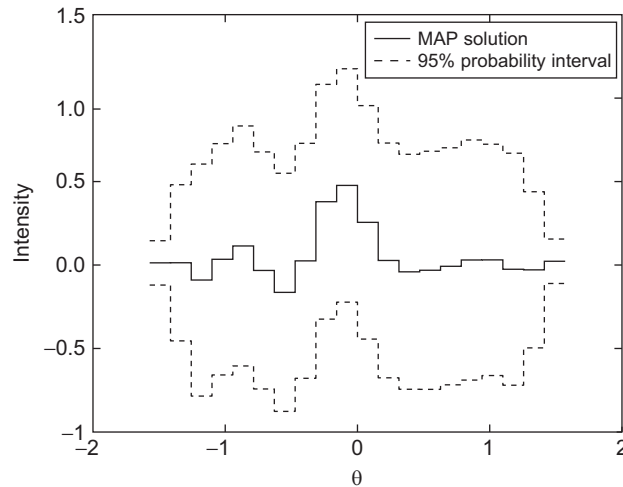


Figure 11.4 The MAP solution of [Figure 11.3](#), with 95% probability intervals.

[Figure 11.5](#) shows a random solution generated from the posterior distribution. This solution varies considerably from the true model, and demonstrates great uncertainty in the inverse solution, consistent with the large 95% probability intervals in [Figure 11.3](#). The roughness of this solution realization is a consequence of the fact that the prior distribution $\mathbf{C}_M$ had zero covariances, so model realizations from the posterior distribution have no preference for smoothness.

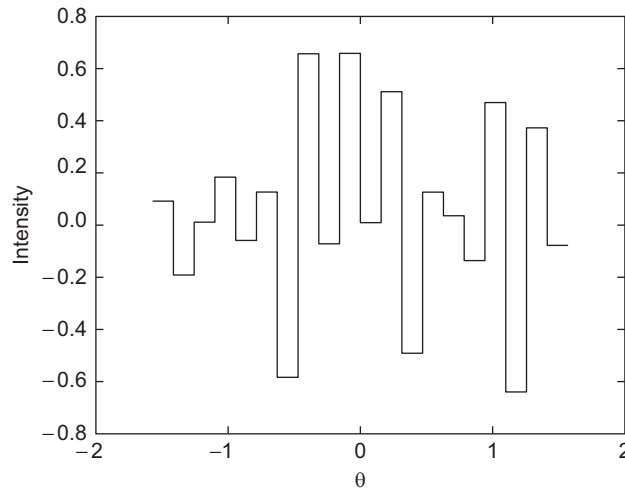


Figure 11.5 A model realization for the Shaw example using an MVN prior distribution with mean 0.5, standard deviation 0.5, and zero covariance.

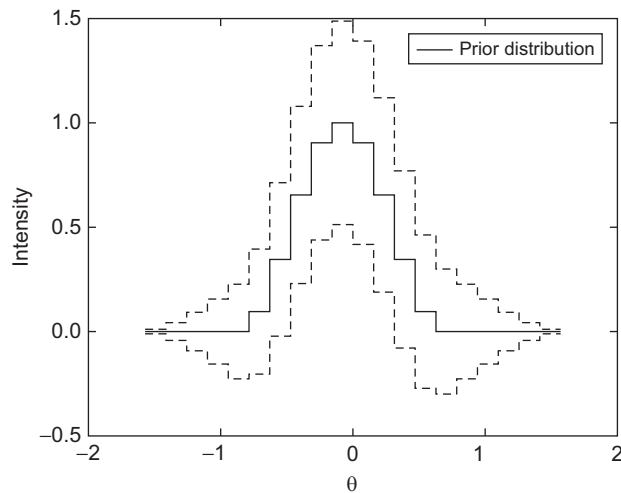


Figure 11.6 A more restrictive zero covariance prior distribution for the Shaw problem with 95% probability intervals.

Next, consider a more restrictive prior distribution. Suppose we have reason to believe that the largest amplitudes in the solution should be near the center of the model. We thus choose the bell-shaped zero-covariance prior distribution depicted in Figure 11.6. Figures 11.7 and 11.8 show the resulting MAP model and its probability

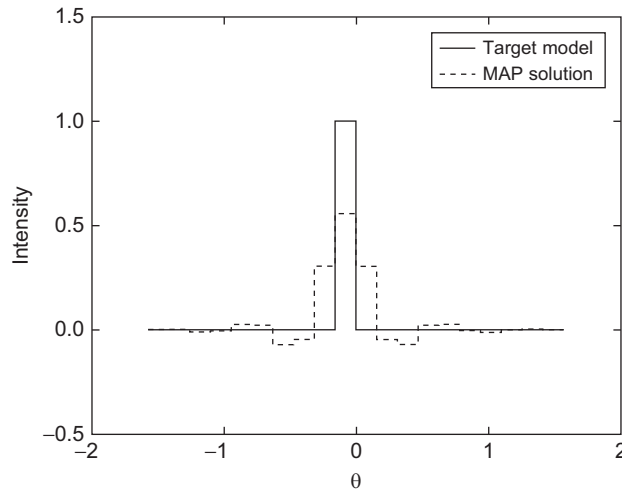


Figure 11.7 The MAP solution for the Shaw example using the prior distribution shown in [Figure 11.6](#).

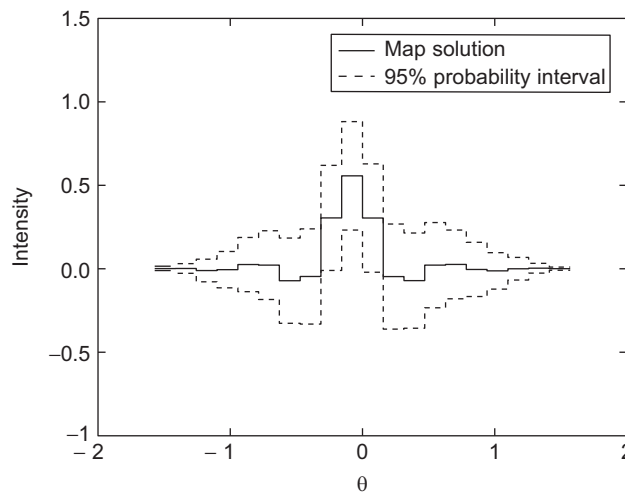


Figure 11.8 The MAP solution of [Figure 11.7](#) with probability intervals.

intervals for this case. The solution recovery is, not surprisingly, improved by our more restrictive prior model given that the true model is highly restricted and is thus consistent with the prior distribution ([Figure 11.3](#)).

These results illustrate a principal issue with applying the Bayesian approach to poorly conditioned problems. To obtain a tight posterior distribution in such cases, we will have to make strong prior assumptions. Conversely, if such assumptions are not made, then

we cannot recover the true model features well. This situation is analogous to that of Tikhonov regularization, which must also impose strong and consistent model bias to produce “good” solutions and must also make strong model assumptions to estimate solution error bounds (Section 4.8).

In the previous example, we applied a prior that preferentially concentrated the model structure in a particular region (the center) by imposing a zero prior with small standard deviations near the model edges. Because this prior distribution had zero covariances, resulting model realizations were rough. Prior distributions can be readily designed to enforce smoothness constraints on realizations from the posterior distribution by specifying a nondiagonal prior covariance matrix. A straightforward way to accomplish this for a 1-dimensional model is to construct a correlation matrix with columns

$$\mathbf{R}_{i,:} = \text{shift}(a_j, i), \quad (11.35)$$

where a_j is the desired sequence of parameter correlations, with a zero lag correlation of 1. The shift operator shifts the sequence (truncating as necessary) so that the zero-lag (unit) maximum of the correlation sequence is centered on element i , and hence on the diagonal of $\mathbf{R}$. Suitable sequences a_i that ensure the positive definiteness of $\mathbf{R}$ can be constructed using autocorrelation (e.g., using the MATLAB `xcorr` function). Here, we use the autocorrelation of a triangle function, which produces a cubic approximation to a Gaussian function. Given the correlation matrix, $\mathbf{R}$, a corresponding prior distribution covariance matrix with uniform parameter variances, σ_M , can then be constructed as

$$\mathbf{C}_M = \sigma_M^2 \mathbf{R}. \quad (11.36)$$

Example 11.3

Consider the vertical seismic profile (VSP) problem first introduced in Example 1.3, which was solved and analyzed using Tikhonov regularization in Examples 4.4, 4.5, 4.6, and 4.7. We revisit this problem as a Bayesian problem, implementing an MVN prior with a covariance matrix of the form of (11.36). The model consists of 50 equally spaced slowness intervals along a 1000-m vertical borehole with an interval length of 20 m. Seismic travel time data are collected at 50 equally spaced depths with independent zero-mean normal errors with standard deviation of $\sigma_D = 2 \times 10^{-4}$ s, producing a data covariance matrix $\mathbf{C}_D = \sigma_D^2 \mathbf{I}$. We apply a prior distribution that is consistent with a seismic slowness decrease (velocity increase) with depth and has a constant gradient between



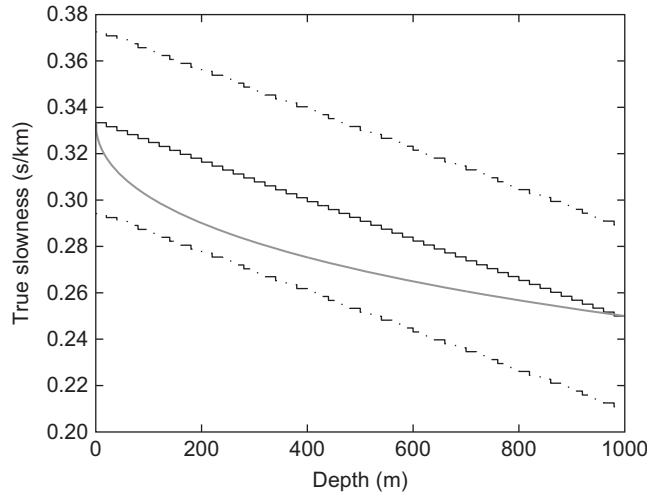


Figure 11.9 A constant-slowness gradient prior distribution and its 95% distribution intervals for the VSP problem. The true model is shown as the gray smooth curve.

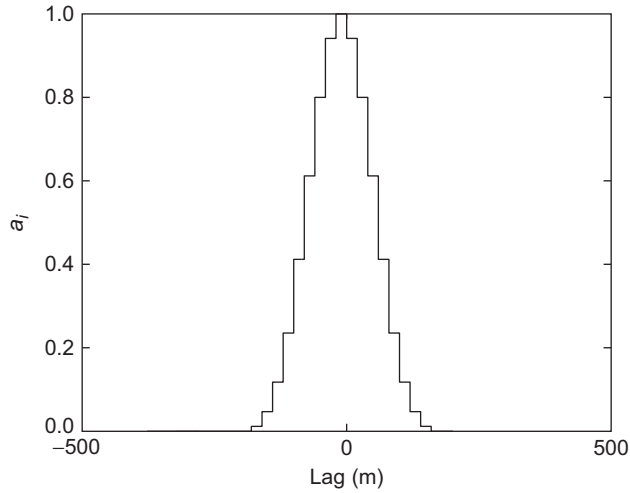


Figure 11.10 A correlation function for the prior distribution of [Figure 11.9](#) with a $1/e$ correlation length of approximately five parameters (100 m).

known seismic slownesses from the top and the bottom of the borehole. We first impose a prior distribution standard deviation of $\sigma_M = 2 \times 10^{-5}$ s/m on all parameters to represent an estimated variability in model slownesses, and utilize a prior correlation function that falls off with a scale length of five model intervals (i.e., a correlation of $1/e$ at a model parameter lag of approximately five model parameters or 100 m). The prior

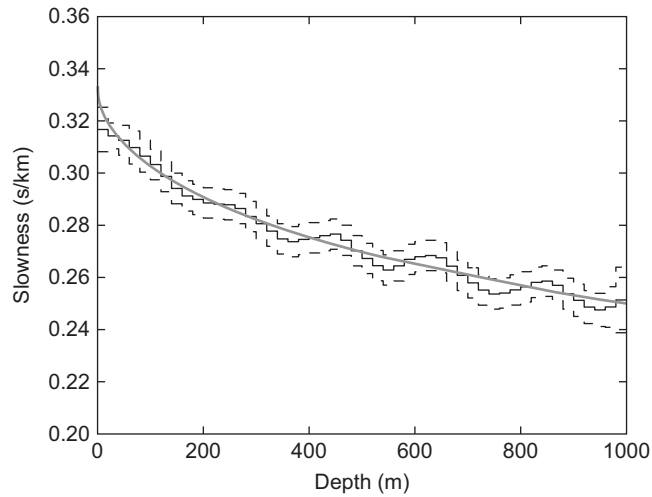


Figure 11.11 MAP model obtained from the posterior distribution, and its 95% distribution intervals, using the prior distribution described in Figures 11.9 and 11.10. The true model is shown as the gray smooth curve.

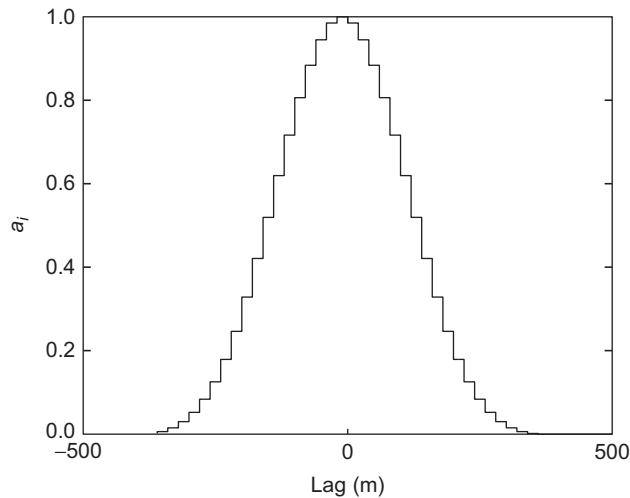


Figure 11.12 A correlation function for the prior distribution of Figure 11.9 with a $1/e$ correlation length of approximately 10 parameters (200 m).

distribution and its parameter standard deviations are shown in Figure 11.9, and the corresponding parameter correlation function is shown in Figure 11.10. The resulting posterior distribution and its standard deviations are shown in Figure 11.11. We next apply a prior with twice the correlation length (Figure 11.12). The resulting posterior distribution is shown in Figure 11.13.

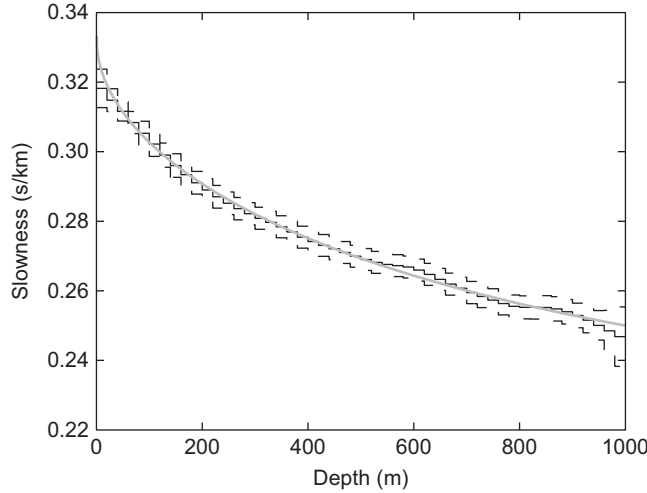


Figure 11.13 MAP model obtained from the posterior distribution, and its 95% distribution intervals, using the prior distribution depicted in Figures 11.9 and 11.12. The true model is shown as the gray smooth curve.

The approach described in this section can be extended to nonlinear problems. To find the MAP solution, solve the nonlinear least squares problem,

$$\min \mathbf{G}(\mathbf{m}) - \mathbf{d})^T \mathbf{C}_D^{-1} (\mathbf{G}(\mathbf{m}) - \mathbf{d}) + (\mathbf{m} - \mathbf{m}_{\text{prior}})^T \mathbf{C}_M^{-1} (\mathbf{m} - \mathbf{m}_{\text{prior}}). \quad (11.37)$$

One can then linearize around the MAP solution to obtain the approximate posterior covariance,

$$\mathbf{C}_{M'} = (\mathbf{J}(\mathbf{m}_{\text{MAP}})^T \mathbf{C}_D^{-1} \mathbf{J}(\mathbf{m}_{\text{MAP}}) + \mathbf{C}_M^{-1})^{-1}, \quad (11.38)$$

where $\mathbf{J}(\mathbf{m})$ is the Jacobian. As with other nonlinear optimization problems, we must consider the possibility of multiple local optima. If (11.37) has multiple solutions with comparable likelihoods, then a single MAP solution and associated $\mathbf{C}_{M'}$ from (11.38) will not accurately characterize the posterior distribution.

11.4. THE MARKOV CHAIN MONTE CARLO METHOD

We next introduce the use of Markov Chain Monte Carlo (MCMC) methods to sample from a posterior distribution. Given a sufficient population of such samples, we can use them to characterize the solution of a Bayesian inverse problem. Because MCMC methods depend only on the forward model and associated likelihood calculations, they are easily applied to both linear and nonlinear problems.

A Markov chain is a sequence of random variables,

$$X^0, X^1, X^2, \dots, \quad (11.39)$$

where the probability distribution of X^{n+1} depends solely on the previous value, X^n , and not on earlier values of random variables in the sequence. That is,

$$P(X^{n+1}|X^0, X^1, \dots, X^n) = P(X^{n+1}|X^n). \quad (11.40)$$

The particular Markov chains considered here will be time-invariant, so that the Markov chain **transition kernel**

$$P(X^n, X^{n+1}) = P(X^{n+1}|X^n) \quad (11.41)$$

is independent of n .

For example, consider a one-dimensional random process in which we start at $X^0 = 0$, and then iteratively compute X^{n+1} from X^n by adding an $N(0, \sigma^2)$ realization to X^n . Clearly X^{n+1} depends directly on X^n , and this dependence is time-invariant. The transition kernel for this Markov chain is

$$k(x_1, x_2) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}(x_1 - x_2)^2/\sigma^2}. \quad (11.42)$$

In the MCMC method, we are interested in Markov chains that have limiting distributions, $q(\mathbf{x})$, such that

$$\lim_{n \rightarrow \infty} P(X^n|X^0) = q(\mathbf{x}). \quad (11.43)$$

Not all Markov chains have limiting distributions. For example, if X^n is the sum of n independent $N(0, \sigma^2)$ random variables, X^n has an $N(0, n\sigma^2)$ distribution. This does not approach a limiting probability distribution as $n \rightarrow \infty$.

For a general multivariate model, it can be shown that if $q(\mathbf{x})$ and $k(\mathbf{x}, \mathbf{y})$ satisfy the **local balance** equation

$$q(\mathbf{x})k(\mathbf{x}, \mathbf{y}) = q(\mathbf{y})k(\mathbf{y}, \mathbf{x}) \quad (11.44)$$

for all models $\mathbf{x}$ and $\mathbf{y}$, then $q(\mathbf{x})$ is the limiting distribution of the Markov chain. Equation (11.44) states that the rate of model transitions from $\mathbf{x}$ to $\mathbf{y}$ equals the rate from $\mathbf{y}$ to $\mathbf{x}$.

The **Metropolis-Hastings sampler** is an algorithm that simulates a Markov chain with a specified limiting distribution. We will apply it to produce samples from a posterior distribution $q(\mathbf{m}|\mathbf{d})$ that will tend to densely sample its higher likelihood regions. With enough such samples, we can usefully characterize the posterior distribution of a Bayesian parameter estimation or inverse problem. Since the data vector $\mathbf{d}$ is given, we will simply write $q(\mathbf{m}|\mathbf{d})$ as $q(\mathbf{m})$ throughout the following development.

We begin a Metropolis–Hastings sampler implementation by picking a **proposal distribution** $r(\mathbf{x}, \mathbf{y})$ that facilitates random steps in the posterior model space. These randomly perturbed samples will subsequently be subjected to a likelihood-based test in the Metropolis–Hastings sampler. A common choice for the proposal distribution is an MVN distribution with zero covariances and variances σ_i^2 , so that

$$r(x_i, y_i) \propto e^{-\frac{1}{2}(x_i - y_i)^2 / \sigma_i^2}. \quad (11.45)$$

Although $r(\mathbf{x}, \mathbf{y})$ cannot be implemented as a transition kernel directly, because it does not satisfy (11.44), this can be remedied by introducing a factor called the **acceptance ratio**,

$$\alpha(\mathbf{x}, \mathbf{y}) = \min(1, s), \quad (11.46)$$

where

$$s = \frac{q(\mathbf{y})r(\mathbf{y}, \mathbf{x})}{q(\mathbf{x})r(\mathbf{x}, \mathbf{y})}. \quad (11.47)$$

Note that $0 \leq \alpha(\mathbf{x}, \mathbf{y}) \leq 1$. Also note that

$$\alpha(\mathbf{y}, \mathbf{x}) = \min(1, s^{-1}). \quad (11.48)$$

At least one of $\alpha(\mathbf{x}, \mathbf{y})$ and $\alpha(\mathbf{y}, \mathbf{x})$ will thus be equal to 1, depending on whether s is greater than or less than 1.

Now, let

$$\tilde{k}(\mathbf{x}, \mathbf{y}) = \alpha(\mathbf{x}, \mathbf{y})r(\mathbf{x}, \mathbf{y}). \quad (11.49)$$

$\alpha(\mathbf{x}, \mathbf{y}) \leq 1$, so

$$\beta(x) = \int_{\text{all models}} \tilde{k}(\mathbf{x}, \mathbf{y}) d\mathbf{y} = \int_{\text{all models}} \alpha(\mathbf{x}, \mathbf{y})r(\mathbf{x}, \mathbf{y}) d\mathbf{y} \leq 1. \quad (11.50)$$

Thus $\tilde{k}(\mathbf{x}, \mathbf{y})$ is not a properly normalized transition kernel. We can, however, obtain a normalized kernel by defining

$$k(\mathbf{x}, \mathbf{y}) = \tilde{k}(\mathbf{x}, \mathbf{y}) + (1 - \beta)\delta(\mathbf{x} - \mathbf{y}) \quad (11.51)$$

so that

$$\begin{aligned} \int_{\text{all models}} k(\mathbf{x}, \mathbf{y}) d\mathbf{y} &= \int_{\text{all models}} \alpha(\mathbf{x}, \mathbf{y})r(\mathbf{x}, \mathbf{y}) d\mathbf{y} + \int_{\text{all models}} (1 - \beta)\delta(\mathbf{x} - \mathbf{y}) d\mathbf{y} \\ &= \beta + (1 - \beta) \\ &= 1. \end{aligned} \quad (11.52)$$

A simple algorithm can now be used to generate a random value $\mathbf{y}$ from $\mathbf{x}$ that satisfies (11.44).

Algorithm 11.1 Transition Kernel Evaluation

1. Generate a candidate $\mathbf{y}$ from $\mathbf{x}$ according to the proposal distribution $r(\mathbf{x}, \mathbf{y})$.
 2. Compute $\alpha(\mathbf{x}, \mathbf{y})$.
 3. With probability α , return the candidate $\mathbf{y}$.
 4. With probability $1 - \alpha$, return the previous value $\mathbf{x}$.
-

Now, we need to show that $q(\mathbf{x})$ and $k(\mathbf{x}, \mathbf{y})$ as defined above satisfy the local balance equation (11.44). If $\mathbf{y} = \mathbf{x}$, this is obviously true. For $\mathbf{y} \neq \mathbf{x}$, we need to consider the two cases of $\alpha(\mathbf{x}, \mathbf{y}) = 1$ or $\alpha(\mathbf{y}, \mathbf{x}) = 1$. If $\alpha(\mathbf{x}, \mathbf{y}) = 1$, then

$$\alpha(\mathbf{y}, \mathbf{x}) = s^{-1} = \frac{q(\mathbf{x})r(\mathbf{x}, \mathbf{y})}{q(\mathbf{y})r(\mathbf{y}, \mathbf{x})} \quad (11.53)$$

and

$$q(\mathbf{x})k(\mathbf{x}, \mathbf{y}) = q(\mathbf{x})\alpha(\mathbf{x}, \mathbf{y})r(\mathbf{x}, \mathbf{y}) = q(\mathbf{x})r(\mathbf{x}, \mathbf{y}). \quad (11.54)$$

Also,

$$q(\mathbf{y})k(\mathbf{y}, \mathbf{x}) = q(\mathbf{y})\alpha(\mathbf{y}, \mathbf{x})r(\mathbf{y}, \mathbf{x}) = q(\mathbf{y})\frac{q(\mathbf{x})r(\mathbf{x}, \mathbf{y})}{q(\mathbf{y})r(\mathbf{y}, \mathbf{x})}r(\mathbf{y}, \mathbf{x}) = q(\mathbf{x})r(\mathbf{x}, \mathbf{y}). \quad (11.55)$$

However, $q(\mathbf{x})r(\mathbf{x}, \mathbf{y}) = q(\mathbf{x})k(\mathbf{x}, \mathbf{y})$, so

$$q(\mathbf{y})k(\mathbf{y}, \mathbf{x}) = q(\mathbf{x})k(\mathbf{x}, \mathbf{y}), \quad (11.56)$$

thus satisfying (11.44). A similar argument shows that (11.44) is satisfied for the case where $\alpha(\mathbf{y}, \mathbf{x}) = 1$.

There are several important tactics that help to simplify the method further. Because the product of q and r appears in both the numerator and denominator of s , we need only know these factors to constants of proportionality, and thus do not need to normalize q and r in individual calculations. Also note that the posterior distribution, $q(\mathbf{m})$, is proportional to the product of the prior, $p(\mathbf{m})$, and the likelihood, $f(\mathbf{d}|\mathbf{m})$ (11.3). We can thus write (11.46) as

$$\alpha(\mathbf{x}, \mathbf{y}) = \min \left(1, \frac{p(\mathbf{y})f(\mathbf{d}|\mathbf{y})r(\mathbf{y}, \mathbf{x})}{p(\mathbf{x})f(\mathbf{d}|\mathbf{x})r(\mathbf{x}, \mathbf{y})} \right). \quad (11.57)$$

If $r(\mathbf{x}, \mathbf{y})$ is a symmetric distribution, such as (11.45), then $r(\mathbf{x}, \mathbf{y}) = r(\mathbf{y}, \mathbf{x})$, and we can simplify (11.57) to

$$\alpha(\mathbf{x}, \mathbf{y}) = \min \left(1, \frac{q(\mathbf{y})}{q(\mathbf{x})} \right). \quad (11.58)$$

In computational practice, numbers in the numerator of (11.58) may be extremely small, and can thus generate floating point underflow problems. This is easily avoided by evaluating $\log \alpha(\mathbf{x}, \mathbf{y})$ instead of $\alpha(\mathbf{x}, \mathbf{y})$. We now have all the components to describe the Metropolis-Hastings sampler.

Algorithm 11.2 The Metropolis-Hastings Sampler

Given a starting model, $\mathbf{m}^0$, repeat the following steps for $j = 1, 2, \dots$ until the posterior distribution is sufficiently sampled by the set of models $\mathbf{m}^j$.

1. Generate a candidate model $\mathbf{c}$ from the previous model, $\mathbf{m}^j$, using the proposal distribution $r(\mathbf{m}^j, \mathbf{c})$.
 2. Compute $\log \alpha(\mathbf{m}^j, \mathbf{c})$.
 3. Let t be a realization of a uniformly distributed random variable on $[0, 1]$.
 4. If $\log t < \log \alpha(\mathbf{m}^j, \mathbf{c})$, then accept the candidate model and let $\mathbf{m}^{j+1} = \mathbf{c}$; otherwise reject the candidate model and let $\mathbf{m}^{j+1} = \mathbf{m}^j$.
-

Note that if $\log t$ is quite small, we will occasionally accept a new model that has a small acceptance ratio and thus move towards a model with reduced likelihood. This property of the algorithm helps overcome the problem of having the sampling become trapped near a localized likelihood maximum. The behavior of the algorithm and its ability to sample the posterior distribution fruitfully and efficiently will depend on the size of the steps taken in generating candidate models. In the case of the normal formulation for the proposal distribution (11.45), this will be controlled by the size of the σ_i . Smaller steps will result in higher acceptance rates, but the algorithm may be unacceptably slow at exploring the posterior distribution. Conversely, larger steps will result in lower acceptance ratios and will thus be less frequently accepted. Either situation may cause the algorithm to become stuck in a particular region of the posterior distribution. Often, the step size parameters are explored adaptively in multiple runs of the algorithm (which may be run as independent parallel processes on a multi-CPU computer system). Some studies, e.g., [51], suggest that the algorithm is optimally tuned when the new model acceptance rate is between approximately 20% and 50%.

11.5. ANALYZING MCMC OUTPUT

Although the limiting distribution of the Markov chain sampled by the Metropolis-Hastings algorithm is the desired posterior distribution, there are significant practical challenges in analyzing the output of an MCMC simulation.

First, successive models $m^k, m^{k+1}, m^{k+2}, \dots$, produced by the simulation are typically strongly correlated with each other, but most statistical techniques require independent samples. In practice, this complication can be avoided by analyzing a

subset of samples that are far apart in the sample sequence. For example, if we examine $\mathbf{m}^k, \mathbf{m}^{k+1000}, \mathbf{m}^{k+2000}, \dots$, it is likely that samples taken 1000 steps apart will not be highly correlated. We can verify this by plotting the successive model autocorrelations for the history of sampling over some moving window length.

Second, early Metropolis-Hastings algorithm samples will be biased by the initial model $\mathbf{m}^0$, which may not lie in a high likelihood region of the posterior distribution. This issue is commonly addressed by skipping over early samples in the chain to give time for the algorithm to “burn in,” and/or by running the process with a variety of starting models, either sequentially or in parallel on multiple CPUs. For example, if it is determined that samples spaced 1000 steps apart are effectively uncorrelated, then it might be reasonable to let the Metropolis-Hastings sampler establish itself for 10,000 steps before beginning to collect samples.

Once we are confident that the procedure has produced a large enough collection of effectively independent samples, we can use the results to characterize the posterior distribution. For a suitably large sample, the MAP solution can be estimated as the retrieved posterior distribution sample that has the greatest likelihood. The posterior distribution may be approximately multivariate normal, which can be established by examining model histograms and quantile-quantile (Q-Q) plots. In this case we can readily construct probability intervals for describing the posterior distribution from the sample mean and covariance using normal assumptions. However, if the distribution is distinctly non-normal, it will be more difficult to produce a simple summary of its character, particularly for very high dimensional models. A common approach is to produce and evaluate scatter plots and/or histograms that display key features of the posterior distribution and to use counting statistics to establish probability intervals.

Example 11.4

Reconsidering the ill-posed nonlinear parameter estimation problem of Example 9.2, we apply the Metropolis-Hastings algorithm to the problem of fitting four parameters, m_i , to the nonlinear function

$$d_i = (G(\mathbf{m}))_i = m_1 e^{m_2 x_i} + m_3 x_i e^{m_4 x_i} \quad (11.59)$$

given a set of observations, d_i , with specified independent normally distributed data noise, specified by corresponding standard deviations σ_i . As in Example 9.2, the true model parameters are $m_1 = 1.0$, $m_2 = -0.5$, $m_3 = 1.0$, and $m_4 = -0.75$; data are produced at 25 equally spaced points, x_i , on the interval $[1, 7]$; and $N(0, 0.01^2)$ independent noise is added to each data element. The likelihood function is specified by

$$f(\mathbf{d}|\mathbf{m}) \propto \prod_{i=1}^m e^{-\frac{1}{2}(d_i - G(\mathbf{m}))_i^2 / \sigma_i^2}, \quad (11.60)$$



which, after taking the natural logarithm, is

$$\log(f(\mathbf{d}|\mathbf{m})) = -\frac{1}{2} \sum_{i=1}^m (d_i - G(\mathbf{m})_i)^2 / \sigma_i^2 + C, \quad (11.61)$$

where C is the logarithm of the constant of proportionality in (11.60). We sample the posterior distribution using 410,000 steps, and produce a low-correlation set of 400 samples by allowing a 10,000 sample burn in and selecting subsequent samples spaced 1000 steps apart by down-sampling the full 410,000 length sequence. The procedure was initiated using a random model selected from a 4-dimensional uniform distribution

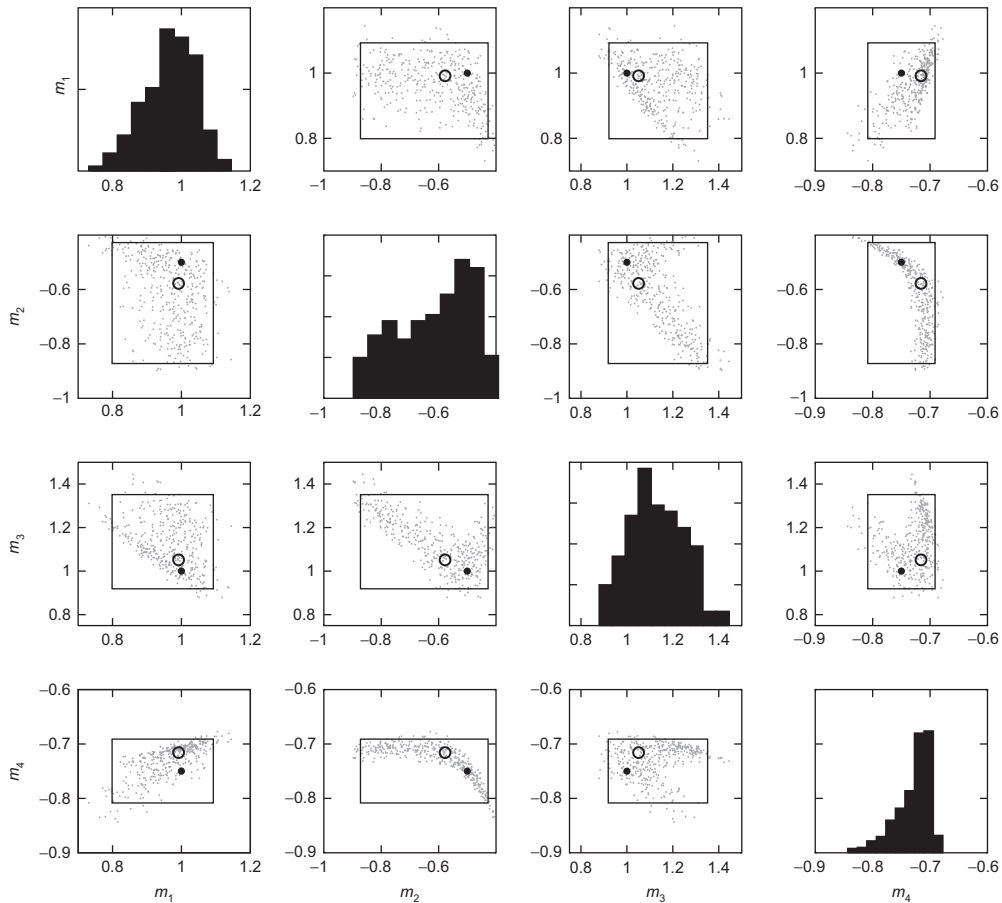


Figure 11.14 Sampled posterior distribution for [Example 11.3](#). The true model is shown as the large black dot, and the MAP model estimated from the maximum likelihood posterior distribution sample is indicated by the open circle. 400 retrieved samples of the posterior distribution (every 1000th calculated sample from the MCMC output) are shown as gray dots. Ninety-five percent probability intervals estimated from the MCMC posterior distribution samples are shown by boxes.

bounded by $[-1, 1]$ in each parameter direction and applying a uniform prior for the region $m_1 = [0, 2]$, $m_2 = [-1, 0]$, $m_3 = [0, 2]$, and $m_4 = [-1, 0]$. In each of the MCMC steps, we apply independent normally distributed random perturbations with standard deviations of 0.005 in each of the four model parameter directions, accepting or rejecting the corresponding candidate models according to the Metropolis-Hastings algorithm. The procedure produced an acceptance rate of approximately 39.5%, which is in the nominally acceptable range of not being too large or too small [51].

Figure 11.14 shows scatter plots and histograms of the resulting sampled posterior distribution, along with the corresponding MAP solution and 95% probability intervals for each parameter direction, which well enclose the true solution. Figure 11.15 shows

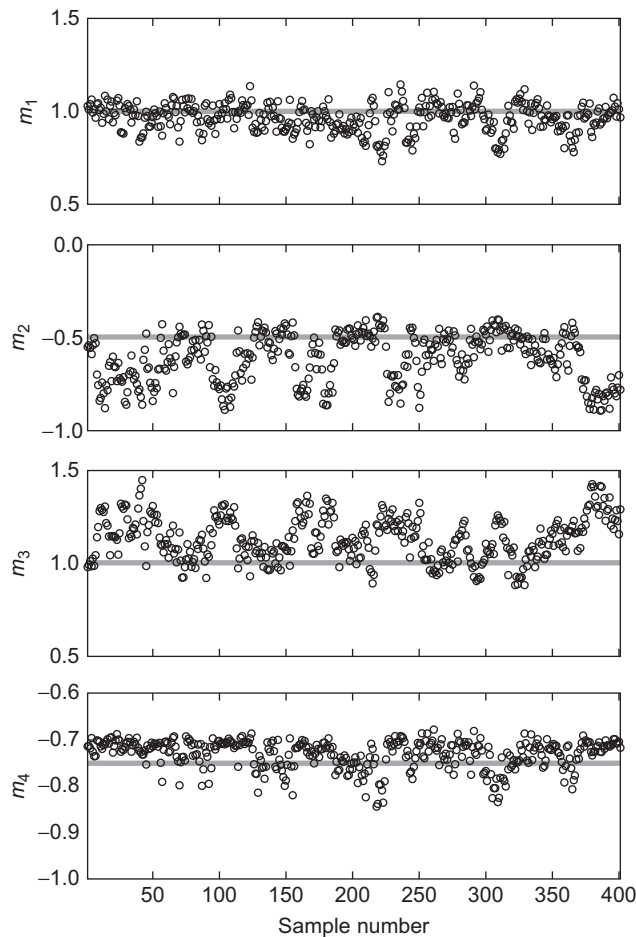


Figure 11.15 Time history of the posterior samples plotted in Figure 11.14. True parameters are shown as gray lines.

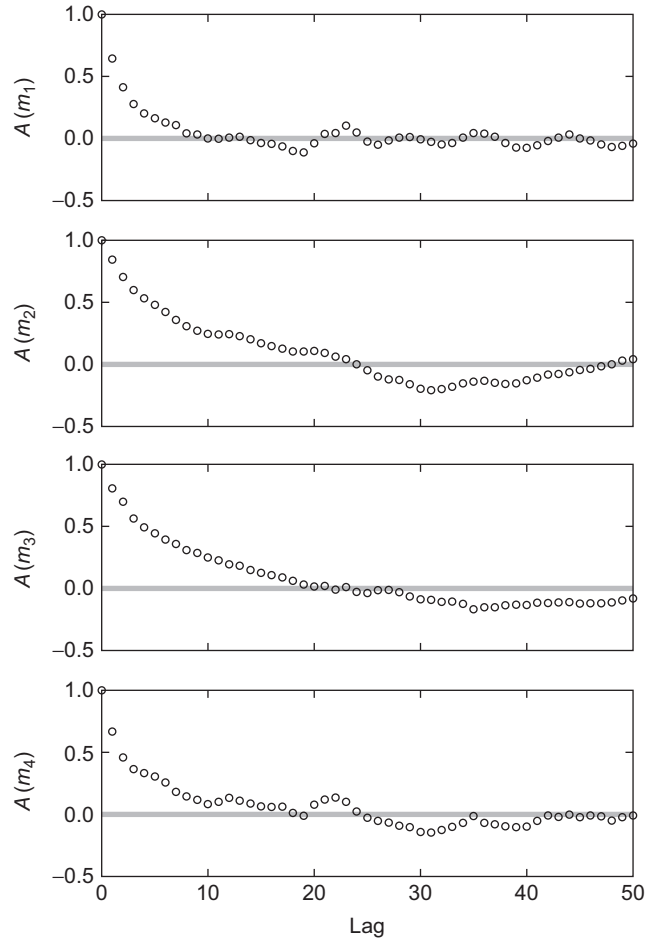


Figure 11.16 Twenty-point autocorrelations for posterior distribution parameters shown in [Figure 11.15](#).

the time history of the corresponding 400 samples. The prominent quasilinear scattering trends in the output between m_2 and m_3 are indicative of a high degree of anticorrelation between some parameter pairs. This anticorrelation is also apparent in the sequence of parameter samples in [Figure 11.15](#), for example in the anticorrelated trends of m_2 and m_3 . An examination of the parameter autocorrelations for the samples shown in [Figure 11.14](#) ([Figure 11.16](#)) shows that individual parameter estimates are effectively decorrelated after 20 or fewer steps. [Figure 11.17](#) shows a histogram of the natural log of likelihood values for the posterior samples. The multimodal nature of this histogram is reflective of the “bumpy” likelihood function surface for this ill-posed problem with random data noise, a situation that also resulted in multiple solutions when we applied

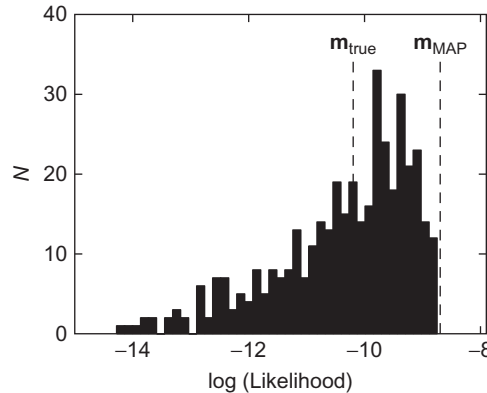


Figure 11.17 Natural logarithm of the likelihood function for the 400 posterior distribution samples shown in Figures 11.14 and 11.15. The value for the true model is approximately -10.2 , and the value for the MAP model is approximately -8.7 .

classical nonlinear solution methods in Example 9.2. However, the MCMC method here provided a sufficiently thorough search of the model space to allow us to discriminate the MAP solution from solutions corresponding to subsidiary likelihood maxima.

11.6. EXERCISES

1. Reanalyze the data in Example 11.1 using a prior distribution that is uniform on the interval $[9, 11]$. Compute the posterior distribution after the first measurement of $10.3 \mu\text{g}$ and after the second measurement of $10.1 \mu\text{g}$. What is the posterior mean?
2. Consider the estimation problem $\mathbf{d} = \mathbf{m}$ (i.e., where $\mathbf{G} = \mathbf{I}$) in two dimensions. The data, $\mathbf{d} = [5 \ 15]^T$, have identical and independent normal errors with standard deviations of $\sqrt{2}$. Apply a zero-mean MVN prior characterized by a covariance matrix where m_1 and m_2 have a correlation coefficient of 0.9 , and equal standard deviations of $\sqrt{5}$. Calculate the MAP model, and compute and plot the 50%, 90%, and 95% contours of the MVN distributions $\mathbf{d}$, $\mathbf{m}_{\text{prior}}$, and the posterior model.
3. In writing (11.28) we made use of the matrix square root.
 - a. Suppose that $\mathbf{A}$ is a symmetric and positive definite matrix. Using the SVD, find an explicit formula for the matrix square root. Your square root should itself be a symmetric and positive definite matrix.
 - b. Show that instead of using the matrix square roots of $\mathbf{C}_D^{-1}$ and $\mathbf{C}_M^{-1}$, we could have used the Cholesky factorizations of $\mathbf{C}_D^{-1}$ and $\mathbf{C}_M^{-1}$ in formulating the least squares problem.
4. Consider the following coin tossing experiment. We repeatedly toss a coin, and each time record whether it comes up heads (0) or tails (1). The bias b of the coin is the

probability that it comes up heads. We have reason to believe that this is not a fair coin, so we will not assume that $b = 1/2$. Instead, we will begin with a uniform prior distribution $p(b) = 1$, for $0 \leq b \leq 1$.

- a. What is $f(d|b)$? Note that the only possible data are 0 and 1, so this distribution will involve delta functions at $d = 0$, and $d = 1$.
- b. Suppose that on our first flip, the coin comes up heads. Compute the posterior distribution $q(b|d_1 = 0)$.
- c. The second, third, fourth, and fifth flips are 1, 1, 1, and 1. Find the posterior distribution $q(b|d_1 = 0, d_2 = 1, d_3 = 1, d_4 = 1, d_5 = 1)$. Plot the posterior distribution.
- d. What is your MAP estimate of the bias?
- e. Now, suppose that you initially felt that the coin was at least close to fair, with

$$p(b) \propto e^{-10(b-0.5)^2} \quad 0 \leq b \leq 1. \quad (11.62)$$

Repeat the analysis of the five coin flips described above.

5. Apply the Bayesian method to Exercise 2 in Chapter 4. Select what you consider to be a reasonable prior. How sensitive is your solution to the prior mean and covariance?
6. Apply the Bayesian method to Exercise 9.3. Assume an MVN prior distribution for α and n , with α and n independent. The prior for α should have a mean of 0.01 and a standard deviation of 0.005, and the prior for n should have a mean of 5 and a standard deviation of 3. Compare your solution with the solution that you obtained to Exercise 9.3. How sensitive is your solution to the choice of the prior?
7. Repeat Exercise 11.6, using MCMC to estimate the posterior distribution. Compare your solution to the solution that you obtained for Exercise 11.6.
8. Apply the Metropolis-Hastings sampler to produce a sampled posterior distribution for the nonlinear parameter estimation problem of Example 9.1. Use a prior distribution that is uniform on $S = [0, 0.01]$ and $T = [0, 2]$, a zero covariance MVN proposal distribution, and a starting model of $(S, T) = (5 \times 10^{-3}, 1.0)$. Generate 200,000 samples using a 10,000-sample burn-in and explore independent step sizes for the two parameters in your proposal distribution to obtain a Metropolis-Hastings sampler acceptance rate between 10% and 50%. Extract every 1000th sample for analysis and establish that these 191 samples are not highly dependent by examining sample autocorrelation functions. Examine the sampled distribution to obtain the MAP model and empirical 95% probability intervals on S and T . Apply a Q-Q plot and assess the multivariate normality of the sampled posterior distribution and compare normal assumption and empirical estimates of the 95% probability intervals.
9. Apply the Metropolis-Hastings sampler to produce a sampled posterior distribution for the nonlinear inverse problem for gravity observations above a buried density



perturbation with an unknown variable depth $m(x)$ and a fixed density perturbation $\Delta\rho$, as described in Exercise 10.2. Your prior should be selected to favor smooth (specified by nonzero parameter correlations) models.

11.7. NOTES AND FURTHER READING

The arguments for and against the use of Bayesian methods in statistics and inverse problems have raged for decades. Some classical references that provide context for these arguments include [35, 42, 80, 81, 136]. Sivia and Skilling's book [143] is a good general introduction to Bayesian ideas for scientists and engineers. The book by Calvetti and Somersalo introduces Bayesian methods for inverse problems including Markov Chain Monte Carlo sampling and includes MATLAB examples [22]. The book by Kaipio and Somersalo provides a more detailed theoretical treatment of Bayesian methods for inverse problems and includes some interesting case studies [83]. An early paper by Tarantola and Valette on the application of the Bayesian approach was quite influential [156], and Tarantola's book is the standard reference work on Bayesian methods for inverse problems [155]. The book by Rodgers [133] focuses on application of the Bayesian approach to problems in atmospheric sounding. The paper by Gouveia and Scales [56] discusses the relative advantages and disadvantages of Bayesian and classical methods for inverse problems. The draft textbook by Scales and Smith [137] takes a Bayesian approach to inverse problems.

In many cases the solution to an inverse problem will be used in making a decision, with measurable consequences for making the "wrong" decision. Statistical decision theory can be helpful in determining the optimal decision. The paper by Evans and Stark provides a good introduction to the application of statistical decision theory to inverse problems [48].